

Face Recognition and Expression Analysis using YOLO11 Architecture

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Abstract. Facial Expression Recognition (FER) is essential for understanding human emotions, improving human-computer interactions, and supporting mental health monitoring. This work presents a novel FER system that uses the YOLO11 architecture to address key challenges, including computational inefficiency, subtle expression detection, and environmental variations such as occlusion and lighting. FER plays a vital role in applications such as healthcare, education, and intelligent systems, bridging gaps in human-computer interaction and enhancing emotional intelligence in machines. The proposed system integrates multiscale detection, feature fusion, and advanced attention mechanisms, such as the C2PSA block, to improve accuracy and robustness. Unlike previous YOLO versions, YOLO11 introduces innovative transformer-based modules and refined attention mechanisms that enhance subtle emotion recognition, particularly under challenging conditions. Using the FER2013 dataset, the system achieved a general mean average precision mAP@50 of 84.0% and mAP@50-95 of 83.9%. High classification accuracy was observed for emotions such as happy and surprise, while lower performance was observed for neutral and sad expressions. These results show that YOLO11 demonstrates improved performance compared to earlier YOLO versions, indicating its suitability for FER in diverse and dynamic conditions. Future work aims to refine subtle emotion detection, expand datasets for greater diversity, and enhance performance for less distinguishable expressions in real-world environments.

Keywords: Attention Mechanisms, Facial Expression Recognition (FER), FER2013 dataset, Real-Time Emotion Recognition, YOLO11.

1 Introduction

Facial Expression Recognition (FER) [1] is essential for understanding human emotions and finds significant applications in areas such as human-computer interaction, mental health evaluation, education, and behavioral studies. Precisely identifying emotions like happiness, sadness, anger, surprise, and neutrality in facial expressions is crucial for developing intelligent systems capable of

responding to human emotions in real time. As the demand for emotion-aware systems increases, achieving high accuracy in recognizing these emotions while maintaining low computational latency becomes a key challenge, particularly in real-world environments where variations in lighting, pose, and occlusion can degrade performance.

Convolutional Neural Networks (CNN) [2] have greatly advanced FER, allowing the extraction of complex facial features that correlate with specific emotions. Traditional FER approaches often rely on multistage pipelines, which require separate models for face detection and emotion classification. However, these methods can suffer from inefficiencies in terms of speed, making them unsuitable for real-time applications. To overcome these challenges, the You Only Look Once (YOLO) [3] architecture provides an integrated framework for real-time object detection, enabling concurrent face detection and emotion classification in a single forward pass. However, many FER systems continue to encounter notable limitations. Current approaches often face challenges in balancing computational efficiency and accuracy, particularly when identifying subtle expressions, such as micro-expressions, or adapting to dynamic scenarios where individuals may wear accessories, such as glasses[4,5]. These challenges underscore the need for an advanced architecture that combines robust expression detection with real-time capabilities, ensuring broader applicability in diverse conditions.

The proposed work utilizes YOLO11, developed by Ultralytics. YOLO11 introduces significant improvements in real-time object detection, offering enhanced accuracy, speed, and efficiency, making it an ideal framework for applications like FER. The YOLO11-based FER system achieved impressive performance on the FER2013 dataset, with an overall mAP@50 of 84.0%. It excelled in classifying emotions like happiness and surprise, but showed lower accuracy for neutral and sad expressions. Compared to YOLOv8n, the proposed YOLO11n demonstrated notable improvements: a 36.00% increase in precision, 8.23% in recall, 15.72% in mAP@50, and 30.52% in mAP@50-95, reinforcing its position as a leading solution for FER under dynamic conditions.

The paper is structured as follows: Section 2 reviews the existing literature, focusing on advancements in FER and models based on YOLO. Section 3 describes the approach, with particular attention to the implementation of YOLO11 for FER applications. Section 4 highlights the experimental outcomes, showcasing the proposed framework’s performance compared to other methods. Finally, Section 5 summarizes the key findings, outlines the contributions, and provides suggestions for future research.

2 Background

FER has advanced considerably with the development of deep learning techniques. Traditional approaches like Local Binary Patterns (LBP) [6] and Histogram of Oriented Gradients (HOG) [7] encountered difficulties in managing real-world variations, including changes in lighting, occlusions, and a wide range of facial expressions. The introduction of CNN allowed automatic feature extrac-

tion, which improved accuracy. However, CNNs were computationally expensive, limiting their use in real-time applications. The YOLO architecture revolutionized FER by combining face detection and expression classification into a single unified model, enabling real-time processing on edge devices. YOLO’s single-pass detection framework reduces computational overhead, making it ideal for real-time FER systems in diverse dynamic environments, bridging the gap between accuracy and efficiency for edge device applications.

Earlier YOLO versions, such as YOLOv3 and YOLOv4, made significant strides in object detection but were limited in handling the complex dynamics of facial expression recognition (FER), especially in real-time settings. YOLOv5 introduced several improvements, but challenges remained in detecting subtle expressions and handling noisy conditions. YOLOv8 [8] marked a milestone in FER by introducing features such as multi-modal fusion and enhanced feature aggregation, which improved the detection of subtle expressions and robustness to occlusions and environmental variations. It incorporated advanced architectural elements, enabling better detection of micro-expressions and handling challenging scenarios such as variable lighting and dynamic movements. However, YOLOv8 faced limitations in detecting extremely subtle expressions in noisy conditions and struggled with fine-grained emotion differentiation.

YOLO11 [9], the latest iteration in the YOLO series, addresses these challenges by introducing significant improvements in architecture and training. YOLO11 incorporates advanced multi-scale feature fusion, refined attention mechanisms [10], and contextual learning enhancements via transformer-based modules and dynamic receptive fields. These innovations allow YOLO11 to detect fine-grained and subtle expressions with high precision, even in complex scenarios involving occlusions, variable lighting, and motion artifacts. The rationale for selecting these specific techniques lies in their ability to adaptively adjust to varying environmental factors and handle overlapping expressions more accurately. By incorporating advanced attention mechanisms, YOLO11 selectively focuses on the most relevant features, enabling better recognition of subtle emotional cues in diverse conditions. Additionally, the transformer-based modules contribute to improved contextual understanding, enabling the model to process facial dynamics more effectively.

The YOLO11 provides an effective solution for FER in dynamic environments, building on the strengths of earlier YOLO versions while overcoming their limitations. Figure 1 illustrates the performance comparison of different versions of YOLO. The subsequent section explores the methodology in detail, focusing on the YOLO11 architecture and its application to FER, highlighting how its innovations address the challenges of previous models.

3 Methodology

The proposed methodology utilizes the advanced YOLO11 architecture to address FER. The model incorporates efficient feature extraction, attention mechanisms, and multi-scale predictions to ensure accurate and real-time expression

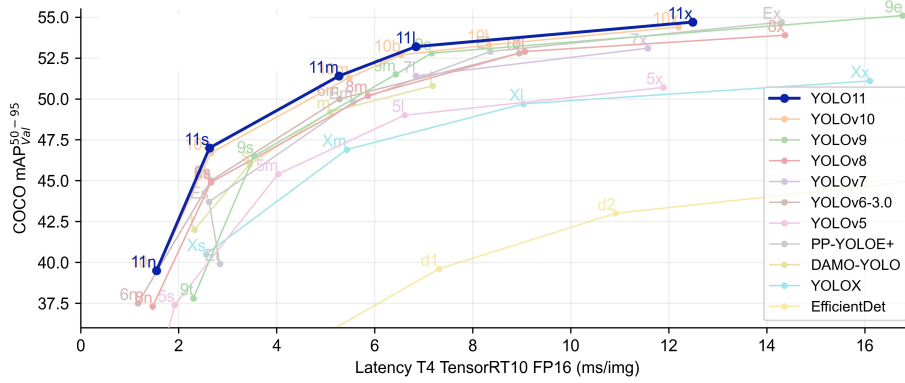


Figure 1. Performance comparison of different versions of YOLO models.[11]

recognition. The approach focuses on optimizing both performance and computational efficiency, making it suitable for diverse and resource-constrained environments.

3.1 Overview of YOLO11 Architecture

FER is a challenging problem in computer vision that requires precise and efficient feature extraction, attention mechanisms, and multiscale predictions. YOLO11 builds on the advancements of previous YOLO architectures to deliver superior performance in FER tasks by introducing components such as the C3K2 block [9], the SPPF module (Spatial Pyramid Pooling Fast) [12], and the C2PSA block (Cross-Stage Partial with Spatial Attention) [9].

Backbone: Efficient Feature Extraction The backbone of YOLO11 is responsible for hierarchical feature extraction. It begins with a Convolutional Block, consisting of a 2D convolution, batch normalization, and the SiLU activation function[13], defined as shown in Equation 1.

$$SiLU(x) = x \cdot \sigma(x), \quad (1)$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$ represents the sigmoid function. The SiLU activation provides smooth gradients, enhancing feature learning in facial emotion recognition tasks.

The Bottleneck Block integrates a residual connection, defined in Equation 2.

$$Y = F(X) + X, \quad (2)$$

where $F(X)$ is the learned transformation and X is the input. This helps mitigate the vanishing gradient problem, crucial for capturing subtle micro-expressions.

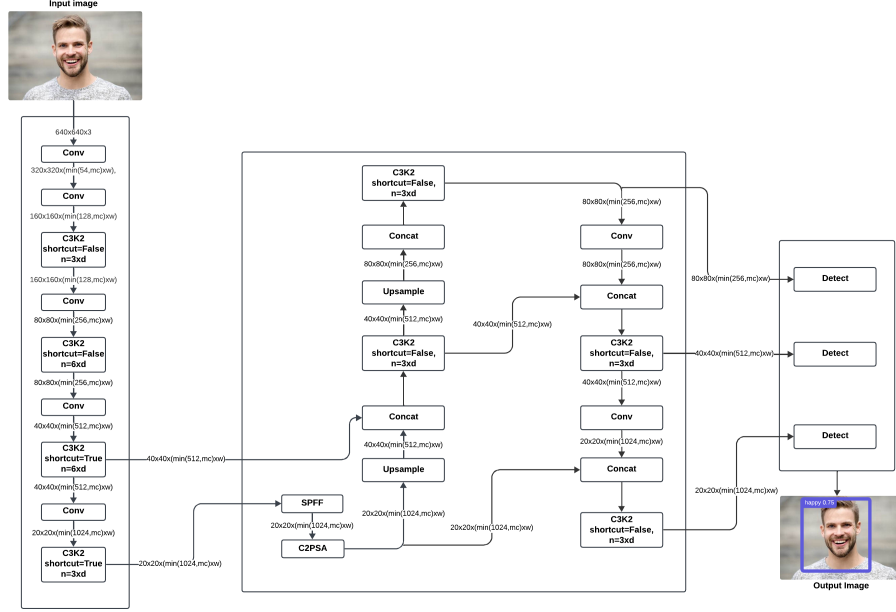


Figure 2. YOLO11 Model Architecture[17]

The C2F Block (Cross-Stage Partial Focus)[14] improves computational efficiency by splitting the feature map into two paths: one processed through bottleneck layers and the other bypassed. The C3K2 Block, a key innovation in YOLO11, uses smaller 3×3 kernels for efficient feature extraction, enhancing local and global features for FER.

Neck: Multi-scale feature aggregate The SPPF Module aggregates spatial information across multiple scales using multiple aggregation layers with varying kernel sizes, ensuring detection of both small and large expressions. The aggregated output is expressed in Equation 3.

$$F_{SPPF} = \text{Concat}(\text{MaxPool}_1(F), \dots, \text{MaxPool}_n(F)), \quad (3)$$

where F denotes the feature map and n indicates the number of cluster scales. This multi-scale aggregation enables reliable FER across different resolutions.

Attention Mechanisms: Focusing on Key Features The C2PSA block introduces spatial attention mechanisms to highlight critical facial regions, such as the eyes and mouth. The spatial attention map is computed in Equation 4.

$$M = \sigma(\text{Conv}_{1 \times 1}(F)), \quad (4)$$

where M represents the attention map, F is the feature map, and σ denotes the sigmoid function. The refined features are computed as outlined in Equation 5.

$$F_{refined} = F \cdot M. \quad (5)$$

This attention mechanism allows the model to concentrate on important regions, enhancing accuracy in situations with occlusion and low lighting.

Head: Multi-Scale Predictions and Classification The detection head outputs predictions at three scales ($P3, P4, P5$) to handle objects of varying sizes. Bounding box regression is performed as shown in Equation 6.

$$x = \sigma(t_x) + a_x, \quad y = \sigma(t_y) + a_y, \quad w = q_w e^{t_w}, \quad h = q_h e^{t_h}, \quad (6)$$

where t_x, t_y, t_w, t_h are the offsets, a_x, a_y are the anchor center coordinates, and q_w, q_h are anchor dimensions [15]. Emotion classification is performed using softmax activation [16] as expressed in Equation 7.

$$P(emotion_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad (7)$$

where z_i is the raw score for the i -th emotion.

YOLO11 has a design that includes efficient backbone processing, multi-scale aggregation, and attention mechanisms, resulting in excellent performance in FER tasks. Figure 2 illustrates the complete structure of the YOLO11 model.

4 Results and Discussion

This section highlights the outcomes of the proposed framework for facial emotion recognition utilizing the YOLO11 architecture. The description of dataset, experimental setup, experimental outcomes, and evaluation metrics are discussed. A detailed comparison of YOLO11 with previous YOLO models highlights its improvements in detection accuracy and emotion recognition performance.

4.1 Dataset Description

In this proposed work, we used the FER2013 dataset [18], a well-established benchmark for FER tasks. The dataset comprises 35,887 labeled grayscale images, each measuring 48×48 pixels. Although the original FER2013 dataset contains seven emotion categories, we focused on five selected emotions for this study: Angry, happy, neutral, sad, and surprise. This decision was made to simplify the emotion classification task and improve the model's performance on the most commonly encountered emotions in real-world scenarios. Figure 3 shows the sample images from the FER2013 dataset.

For consistency and to ensure compatibility with the YOLO11 model, all images in the dataset were pre-processed. Specifically, they were resized to a uniform resolution of 256×256 pixels to retain facial details while optimizing computational efficiency. Furthermore, all pixel values were normalized to a standard range of $[0, 1]$, enhancing the stability of the training process and ensuring consistent input to the neural network.



Figure 3. Sample Images from the FER2013 Dataset

4.2 Experimental Setup for Training

All experiments were conducted on a GPU-accelerated system optimized for deep learning applications. The setup utilized NVIDIA GPUs to provide the computational power required for training advanced deep learning models such as YOLO11. The Ultralytics framework, integrated with PyTorch, was used for its flexibility and scalability, enabling efficient training of the YOLO11 model on a custom emotion detection dataset. Early stopping mechanisms ensured optimal training efficiency and the best performing model was saved for deployment.

4.3 Training and validation

In this proposed work, we trained the YOLO11n model using the FER2013 dataset. The model was downloaded from the Ultralytics repository [11]. For training, we used 1,000 images for each of the five selected emotions: Angry, Happy, Neutral, Sad and Surprise, ensuring that the data set remained balanced

and preventing overfitting. The model was trained using the Adam Optimizer [19] with an initial learning rate of 0.001, which was adjusted dynamically throughout the training process. The training lasted for 50 epochs, with a batch size of 16, facilitating efficient convergence of the model.

To assess the model's performance during training and validation, various evaluation metrics were employed, including precision, recall, mAP@50, and mAP@50-95. These metrics are crucial for measuring the model's accuracy and its ability to localize facial expressions.

4.4 Experimental Outcomes

Figure 4 shows the performance metrics for training and validation of the YOLO11n model on the FER2013 dataset for facial emotion recognition. The training losses, including box_loss, cls_loss, and dfl_loss, consistently decrease over time, reflecting enhancements in the model's ability to both localize and classify facial expressions. Similarly, the validation losses follow a downward trend with minor fluctuations, reflecting effective generalization to the validation dataset. Precision and recall metrics improve steadily over epochs, stabilizing at high values. The mAP@50 and mAP@50-95 demonstrates strong overall performance.

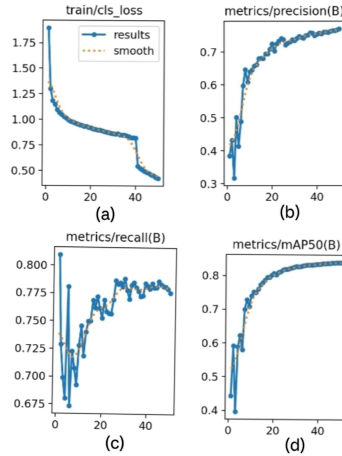


Figure 4. Training and Validation Metrics of the YOLO11n Model (a) Training box loss graph indicating the reduction in error for bounding box predictions as the model learns over epochs. (b) Precision graph showing the model's ability to minimize false positive predictions during training. (c) Recall graph illustrating the model's effectiveness in identifying true positive detections over epochs. (d) mAP50 graph depicting the model's overall performance in balancing precision and recall.

Table 1 presents the evaluation metrics for each emotion class, highlighting the performance of the model in terms of precision, recall, mAP@50, and mAP@50-95. The emotions Happy and Surprise achieved the highest scores on all metrics, demonstrating the model’s strong ability to detect and classify these emotions accurately. In contrast, Neutral and Sad exhibited lower scores, particularly in the recall and mAP metrics, indicating areas where the model’s performance could be improved. The disparity in performance across emotion categories is due to the inherent difficulty of distinguishing subtle emotional expressions, and the lower number of instances in the dataset for emotions like Sad and Neutral. The lower recall and mAP values for these emotions could be attributed to their often subtle and less expressive nature, which makes them more challenging for the model to detect.

Table 1: Evaluation Metrics for Each Emotion Class

Emotion	Precision	Recall	mAP@50	mAP@50-95
Angry	0.744	0.691	0.796	0.794
Happy	0.884	0.9	0.959	0.958
Neutral	0.659	0.756	0.761	0.761
Sad	0.714	0.637	0.745	0.744
Surprise	0.856	0.894	0.939	0.939
Overall Average	0.771	0.776	0.84	0.839

The overall average metrics, with a precision of 0.771, recall of 0.776, mAP@50 of 0.84, and mAP@50-95 of 0.839, reflect the model’s balanced yet slightly uneven performance, with room for optimization in challenging emotion classes. Figure 5 visually contrasts the ground truth labels with the predictions of the YOLO11n model, further highlighting the accuracy of the model and areas where its classification and detection capabilities can be enhanced.

The YOLO11n model demonstrates notable improvements in accuracy and reliability compared to previous YOLO models, establishing a benchmark for FER. As shown in Table 2, YOLO11n excels in both precision (0.771) and recall (0.776), surpassing earlier models such as YOLOv9, YOLOv8, YOLOv7, and YOLOv5, which had lower scores in these metrics. With an mAP@50-95 of 0.839 and an mAP@50 of 0.84, YOLO11n effectively captures complex facial expressions, addressing the limitations of lightweight models that frequently compromise accuracy for speed. This model demonstrates substantial performance improvements, making it a more reliable and efficient choice for FER tasks. By striking an optimal balance between accuracy and computational efficiency, YOLO11n offers a powerful solution for real-world FER applications.

While YOLO11n demonstrates strong performance overall, certain areas require further attention to enhance detection accuracy, particularly for challenging expressions like Neutral and Sad. One potential avenue for improvement is augmenting the dataset with additional examples of subtle expressions, especially for these underperforming categories. Moreover, refining the attention



Figure 5. Comparison of ground truth labels and YOLO11n model predictions

mechanisms and incorporating additional contextual information, such as facial landmarks or pose estimation, could help the model better distinguish between similar facial expressions. By addressing these limitations, future iterations of the YOLO architecture can achieve even higher levels of performance in complex FER tasks.

Table 2: Evaluation Metrics for Each Emotion Class[20]

Model	Precision	Recall	mAP@50	mAP@50-95
YOLOv5s	0.593	0.718	0.719	0.683
YOLOv7	0.460	0.725	0.558	0.558
YOLOv8n	0.567	0.717	0.726	0.643
YOLOv9c	0.677	0.811	0.730	0.572
YOLO11n	0.771	0.776	0.84	0.839

5 Conclusion and Future Scope

The proposed YOLO11n model demonstrates significant progress in FER, achieving high precision, recall, and mAP scores for most emotions, particularly for distinct expressions like Happy and Surprise. However, challenges remain in accurately recognizing subtler emotions such as Sad and Neutral, highlighting the

need for improved handling of nuanced expressions. The model's robustness and efficiency make it well-suited for real-time applications, such as healthcare diagnostics, customer interaction systems, and video-based emotion analysis, where reliable FER systems are critical. Future efforts should focus on addressing imbalances in training data and enhancing the detection of subtle emotions through advanced techniques like domain adaptation, fine-grained feature extraction, and improved attention mechanisms. Expanding the dataset to include a broader range of emotional expressions and ensuring demographic and cultural diversity can enhance the model's generalizability. Additionally, optimizing the model for deployment on edge devices, refining it to handle environmental challenges like lighting and occlusions, and integrating temporal dynamics for emotion recognition in video sequences are promising avenues for future research. These steps will further establish a practical and effective FER solution for dynamic real-world scenarios.

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